

WorldQuant BRAIN Alpha Mining

AI-Driven Systematic Factor Discovery

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Core Innovation: Built a Claude Code SKILL system that autonomously mines stock alpha factors on the WorldQuant BRAIN platform — from expression generation to submission, polling, and knowledge base updating — in a fully automated loop.

1. MOTIVATION

WorldQuant BRAIN is a global quantitative finance competition where participants submit alpha expressions (FASTEXPR) evaluated against historical market data. Traditional manual mining is labor-intensive: write expression → submit → wait → check results → repeat. Each iteration takes 2–5 minutes of human attention. This project asks: *can an AI agent automate the entire loop, systematically explore the strategy space, and self-improve from failures?*

2. SYSTEM ARCHITECTURE

The mining pipeline consists of four components orchestrated by a Claude Code SKILL:

COMPONENT	TECHNOLOGY	FUNCTION
SKILL Entry	Claude Code SKILL.md	Defines the agent's persona, strategy library, and decision logic for expression generation
API Layer	Python · <code>brain_api.py</code> (7 KB)	Handles BRAIN authentication, alpha submission, result polling, and structured output
Evolution Engine	Python · <code>evolve_skill.py</code> (7.4 KB)	Analyzes results, updates knowledge base, and guides strategy selection for subsequent rounds
Knowledge Base	Markdown (4 files)	Operator syntax rules, valid field inventory, failure diagnosis, successful alpha patterns

3. MINING EXECUTION

On June 24, 2026, the system ran **55 rounds in ~1.5 hours** (20:25–21:52), testing **6 distinct strategy directions**:

STRATEGY	ROUNDS	CLEAN ALPHAS	OUTCOME
<code>ts_rank</code> + fundamental ratios	20+	2	✓ Only viable direction
<code>ts_zscore</code> (price/momentum)	8	0	✗ Negative returns, high turnover
<code>group_rank</code> (cross-sectional)	6	0	✗ Sharpe consistently < 0.8
<code>ts_corr</code> (correlation)	6	0	✗ Near-zero predictive power
Nested <code>group(ts_rank)</code>	6	0	✗ Moderate, no clean pass
Pure momentum / reversal	4	0	✗ Turnover > 75%

4. RESULTS

ALPHA ID	EXPRESSION	SHARPE	FITNESS	TURNOVER	SETTINGS
xAxcdYPq	<code>ts_rank(cashflow_op / cap, 60)</code>	1.47	1.14	0.10	decay=10, SUBINDUSTRY, TOP3000
9qwr2Qz1	<code>ts_rank(cashflow_op / cap, 40)</code>	1.29	1.02	0.12	decay=10, INDUSTRY, TOP3000

Field Efficacy Ranking

EXPRESSION PATTERN	BEST SHARPE	RATING
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<code>cashflow_op / cap</code>	1.47	 Champion
<code>operating_income / equity</code>	1.59	 High Sharpe but Fitness consistently below threshold
<code>operating_income / cap</code>	1.22	 Moderate
<code>cashflow_op / equity</code>	1.12	 Underperforms

5. KNOWLEDGE DISTILLATION

The 55-round process produced cumulative, reusable knowledge:

- **5 FASTEXPR Iron Rules:** (1) No arithmetic between operator results. (2) No cross-family operator mixing. (3) Arithmetic must be inside operators. (4) `group_zscore` does not exist; use `group_rank`. (5) Negation (`-1 *`) is the only allowed operator chaining.
- **Valid Fields:** `operating_income`, `cashflow_op`, `equity`, `cap`, `returns`, `close`. Not valid as group arguments: `gross_margin`, `book_to_price`, `earnings_yield`, `roe`.
- **Neutralization Effect:** INDUSTRY outperforms SUBINDUSTRY for shorter windows (40d); SUBINDUSTRY is superior for longer windows (60d).

6. TECHNICAL STACK

Python

Claude Code SKILL

WorldQuant BRAIN API

FASTEXPR

JSONL Logging

Automated Polling

Knowledge Base

7. SIGNIFICANCE

This project demonstrates three capabilities relevant to quantitative finance research:

- **AI-Finance Integration:** Practical deployment of an LLM-based agent system for systematic alpha discovery — bridging natural language reasoning with quantitative strategy generation.
- **Engineering Discipline:** Full pipeline from API integration to result persistence to knowledge management, with 11 archived iterations documenting the evolution from prototype to production.
- **Research Rigor:** Systematic exploration of 6 strategy directions with documented null results — not just cherry-picking wins, but building an evidence base of what does and does not work.

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